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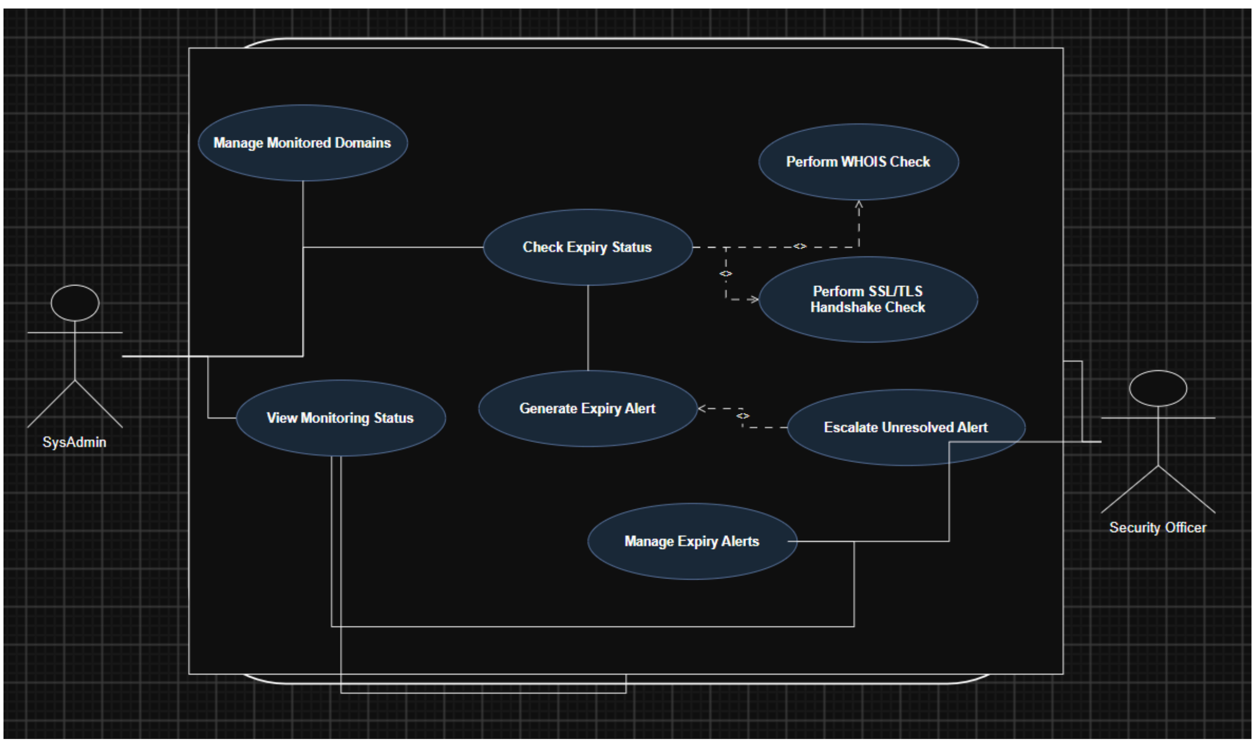
Problem Statement: Domain SSL Expiry Alert
Problem Number: 47

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Perform daily SSL/TLS handshake checks and retrieve certificate expiry dates.	High	Handshake succeeds and expiry date is retrieved; failures are recorded and alerted.	Ensures SSL/TLS certificates are functioning and valid
FR-002	Functional	Perform daily WHOIS checks to retrieve domain registration and expiry information.	High	WHOIS data and expiry date are retrieved successfully; failures are logged	Prevents unexpected domain expiration
FR-003	Functional	Identify domains and certificates approaching expiry using configured warning periods	High	Items within warning thresholds are correctly identified.	Enables preventive action before expiration.
FR-004	Functional	Generate and send expiry alerts to the appropriate SysAdmin or Security Officer	High	Alert contains domain, expiry date and severity and is delivered correctly	Ensures responsible personnel are notified.
FR-005	Functional	Escalate unresolved expiry alerts according to a configured escalation schedule	Medium	Unresolved alerts are escalated after the defined threshold.	Prevents ignored alerts from causing service disruption

NFR-001	Non Functional	Scan 1,000 domains and SSL/TLS endpoints within 3 minutes while maintaining security.	High	Benchmark confirms completion within 3 minutes under simulated peak load.	Ensures scalability and timely monitoring
NFR-002	Non Functional	Maintain at least 99.5% monthly availability for monitoring and alert generation.	High	Monthly availability is ≥99.5%.	Prevents missed monitoring checks and alerts.



Use-Case Flow Specification

Primary Actors: SysAdmin, Security Officer
System: Domain & SSL Certificate Expiry Alert System

1. Preconditions

1. The monitoring system is operational.
2. At least one domain is registered in the system for monitoring.
3. WHOIS services and SSL/TLS endpoints are accessible.
4. Warning thresholds for domain and SSL/TLS certificate expiry have been configured.

2. Postconditions

1. Domain registration and SSL/TLS certificate information is updated.
2. The expiry status of monitored domains and certificates is determined.
3. Monitoring results are stored in the system.
4. Alerts are generated when configured expiry thresholds are reached.
5. The appropriate SysAdmin or Security Officer is notified when an alert is generated.

3. Main Success Scenario

1. The system starts the scheduled daily monitoring process.
2. The system retrieves the list of monitored domains.
3. The system performs a WHOIS lookup for each monitored domain.
4. The system retrieves the domain registration expiry date.
5. The system establishes an SSL/TLS handshake with the monitored endpoint.
6. The system retrieves the SSL/TLS certificate expiry date.
7. The system compares the retrieved expiry dates with the configured warning thresholds.
8. The system determines the expiry status of each monitored domain and SSL/TLS certificate.
9. The system stores the monitoring results.
10. If an expiry threshold has been reached, the system generates an expiry alert.
11. The system sends the alert to the responsible SysAdmin or Security Officer.
12. The system completes the monitoring process.

4. Alternate Flow – SSL/TLS Handshake Failure

1. During the SSL/TLS check, the system fails to establish a secure handshake with the monitored endpoint.
2. The system records the handshake failure.
3. The system marks the endpoint as requiring attention.
4. The system generates a security monitoring alert.
5. The system sends the alert to the responsible Security Officer.
6. The system continues monitoring the remaining domains.

5. Special Requirements

- The monitoring process shall run automatically on a daily schedule.

- Alerts shall include the affected domain, expiry date or failure information, and alert severity.
- The system shall support configurable expiry warning thresholds and escalation rules.